

Yingjie Zhang

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Education

Duke University, M.S. in Computer Science, GPA: 3.67/4.0 Aug. 2025 – May 2027
Shanghai Jiao Tong University, B.Eng. in Computer Science, GPA: 87.41/100 Sept. 2021 – June 2025

Internship Experience

Machine Learning Engineer Intern, TikTok – San Jose, CA May – Aug. 2026

- Developed a **DIN**-based click-sequence module for ads identity matching; improved offline AUC by **6.36%** and High-Accuracy Match Rate (HAMR) by **3.46pp** with flat-to-positive conversion coverage.
- Enhanced the online **MMoE** model by enriching LHUC input with vetted dense user/event features; launched experiments that lifted AUC by **0.14%**, HAMR by **1.67pp**, and conversion coverage by **1.62%**.
- Diagnosed **label leakage** in an upstream candidate-reuse filter; traced a **20.39pp HAMR drop** through stage- and channel-level analysis, used linear probes to identify weak representations in its dedicated **MoE expert**, and communicated the findings to support model improvements.

Software Engineer Intern, Alibaba Group (Amap) – Xiamen, China May – Aug. 2025

- Built a **multi-agent** map theme generation system with **AutoGen Swarm**-based orchestration, integrating **multimodal LLMs** (Gemini, GPT-4); placed **3rd** among 30+ teams in the Agent Show competition.
- Designed a high-level intent-recognition agent to route user requests across downstream agents, and implemented a **TypeScript/React** color-palette workflow with **AutoGen Studio** for real-time generation control.

Machine Learning Engineer Intern, Meituan – Shanghai, China June – Sept. 2024

- Implemented a knowledge-distillation pipeline using **SFT** and custom labeling to compress the 120B **Longcat-Prime** model into 7B **Qwen2-7B-Instruct** for efficient deployment.
- Developed a multilingual merchant-review matching system using **SQL**-extracted data, improving precision by **6.14pp** (91.2% to 97.34%) over the regex-based solution while supporting **12K+** daily model calls.

Research & Projects

Long-Horizon LLM Agent Post-Training for ARC-AGI-3 🔄 GitHub Jan. – May 2026

- Established zero-shot **RLM** and multi-turn **GRPO** pipelines for **ARC-AGI-3**, integrating stateful **SkyRL** rollouts, FSDP training, and vLLM inference for Qwen2.5-3B/Qwen3-8B on four NVIDIA A6000 GPUs.
- Designed frame-diff observations and rollout visualization; ablations across reward shaping, oracle-guided warm-up, and **KL regularization** exposed repetitive-click reward hacking and a persistent gap between reward gains and level completion.

Large-Scale EEG Models for Meditation State Recognition 🔄 Paper — ICASSP 2026 Mar. – Aug. 2025

- As **first author**, designed and conducted a systematic **benchmark** of **large-scale EEG models** with different **self-supervised pretraining strategies** across multiple downstream recognition settings.
- Compared five representative **Transformer**-based models (EEGPT, BENDR, BIOT, LaBraM, Gram), uncovering model-task alignment effects and performance degradation under mismatched continued pretraining.

Addressing Temporal and Auditory Factors in Meditative EEG with Self-Supervised Learning 🔄 Paper — BIBM 2024 Mar. – Sept. 2024

- As **co-first author**, adapted and implemented the **Multi-view Spectral-Spatial-Temporal Masked Autoencoder** (MV-SSTMA) for meditative EEG, outperforming seven baselines by **6.49pp** F1 in subject-dependent and **5.86pp** in cross-subject evaluations.

Skills

Languages: Python, C/C++, SQL, TypeScript

Systems & Deployment: FastAPI, Docker, Linux, Git

ML/DL: PyTorch, TensorFlow, Hugging Face, FSDP

ML Methods: Ranking, Mixture-of-Experts

LLM/NLP: GRPO, SFT, Knowledge Distillation, vLLM, AutoGen